

Fraud Amount Pattern Detection Policy

SentinelIQ Internal Policy Document - Confidential

1. Purpose and Scope

This policy defines detection rules and investigation procedures for amount-based fraud patterns. Amount patterns include round amounts, split transfers, structuring, and threshold-avoidance schemes. These patterns are commonly used to evade automated detection systems that trigger on specific monetary thresholds. This policy applies to all transactions processed by SentinelIQ across HDFC, SBI, ICICI, and Axis bank networks.

2. Definitions

Round Amount: A transaction where the amount is an exact multiple of 500 INR and is greater than or equal to 10,000 INR. Split Transfer: A pattern where a large sum is divided into multiple smaller transactions to avoid detection thresholds, typically executed within a short time window. Structuring: Deliberately breaking transactions into amounts below reporting thresholds (typically 50,000 INR for Indian banking regulations). Threshold Avoidance: Transactions deliberately set just below system-defined or regulatory alert thresholds. Dormant Account Reactivation: A previously inactive account suddenly showing large or unusual transaction activity.

3. Detection Rules -- Round Amount

Rule A-1: A transaction triggers the `round_amount_flag` when BOTH conditions are met: (a) amount modulo 500 equals zero, AND (b) amount is greater than or equal to 10,000 INR. Rule A-2: Round amounts below 10,000 INR are excluded to reduce false positives from normal retail transactions (e.g., 500, 1000, 2000 INR are common legitimate payment amounts). Rule A-3: When `round_amount_flag` is combined with `velocity_flag`, the combined pattern is classified as a potential automated drain attempt -- repeated withdrawals of clean round amounts indicate scripted exploitation. Rule A-4: Round amounts of exactly 50,000 INR or above receive additional scrutiny as they approach or exceed regulatory reporting thresholds.

4. Detection Rules -- Split Transfer Pattern

Rule A-5: A split transfer pattern is identified when 3 or more transactions from the same account, within a 2-hour window, are sent to different `merchant_ids`, and the individual amounts are between 5,000-15,000 INR each but sum to more than 40,000 INR. Rule A-6: Split transfers across different banks (cross-bank splitting) are given higher risk weight as they indicate deliberate evasion of single-bank monitoring systems. Rule A-7: Split transfer detection interacts with `velocity_flag` -- accounts showing both patterns simultaneously are escalated to HIGH tier minimum.

5. Detection Rules -- Dormant Account Reactivation

Rule A-8: An account is considered dormant if it has had zero transactions in the preceding 30 days. The `new_merchant_flag` partially captures this when a dormant account transacts with any merchant. Rule A-9: Dormant accounts reactivating with round amounts above 25,000 INR are flagged with both `new_merchant_flag` and `round_amount_flag`, producing a minimum base score of 40 (2 flags x 20 points). Rule A-10: Multiple dormant accounts reactivating within the same time window and transacting with overlapping merchants indicates potential merchant collusion and should be cross-referenced in the network graph.

6. Thresholds and Scoring Impact

The `round_amount_flag` contributes 20 points to the base risk score. Combined scoring examples: - Round amount only: 1 flag x 20 = 20 (LOW tier, auto-clear) - Round amount + unusual hour: 2 flags x 20 = 40, then x 1.3 = 52 (MED) - Round amount + velocity + unusual hour: 3 x 20 = 60, x 1.3 = 78 (HIGH) - Round amount + velocity + geo + unusual hour: 4 x 20 = 80, x 1.3 = 104, clamped to 100 (CRITICAL) - All 5 flags: 5 x 20 = 100, x 1.3 = 130, clamped to 100 (CRITICAL) The clamping to [0, 100] ensures no score exceeds the maximum regardless of flag combinations. The night multiplier (1.3x) is only applied when `unusual_hour_flag` is True.

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7. Investigation Procedures

Step 1: For round amount flags, verify whether the merchant typically receives round-amount payments (e.g., utility companies, insurance premiums). Common legitimate round amounts include rent payments and EMI instalments. Step 2: For suspected split transfers, aggregate all transactions from the account within the 2-hour window. Calculate total outflow and identify recipient merchant distribution. Step 3: Cross-reference amounts with historical transaction patterns for the account. A first-time 50,000 INR transaction from a typically low-value account is more suspicious than the same amount from a high-net-worth account. Step 4: Check for structuring indicators -- transactions at 49,900, 49,500, or similar amounts just below 50,000 INR reporting threshold. Step 5: For dormant account reactivation, verify account ownership status and check for recent credential changes or new device logins. Step 6: Document all findings in the PDF investigation report under the appropriate tier section. Include amount analysis charts if multiple related transactions are identified.

8. Merchant Collusion Indicators

Merchant collusion is a pattern where one or more merchants knowingly participate in fraudulent transaction schemes. Amount patterns provide key indicators: 1. Multiple accounts sending identical round amounts to the same merchant within a short time window. 2. A merchant receiving transactions exclusively in round amounts above 10,000 INR from accounts that previously had no relationship. 3. Circular patterns where funds flow account -> merchant -> account in closed loops, detectable via NetworkX cycle analysis. The network graph on Page 2 highlights merchant nodes with suspicious incoming transaction patterns. Merchants with high in-degree from new accounts combined with round amounts are tagged for investigation. The Pattern Store records analyst decisions on merchant-related patterns to refine future scoring.

9. Regulatory Compliance

Indian banking regulations require suspicious transaction reporting (STR) for transactions exceeding certain thresholds. SentinelIQ assists compliance by: 1. Automatically flagging transactions at or above 50,000 INR with additional risk factors. 2. Detecting structuring attempts designed to evade reporting. 3. Generating audit-ready PDF reports with full transaction details. 4. Maintaining pattern memory to track recurring suspicious patterns. All amount-pattern detections are logged with the transaction_id, amount, flag combination, and resulting tier for audit trail purposes. Reports are uploaded to S3 (fraud-investigation-group9) and retained per the data retention policy.